

Module 6

Implementing Network Services

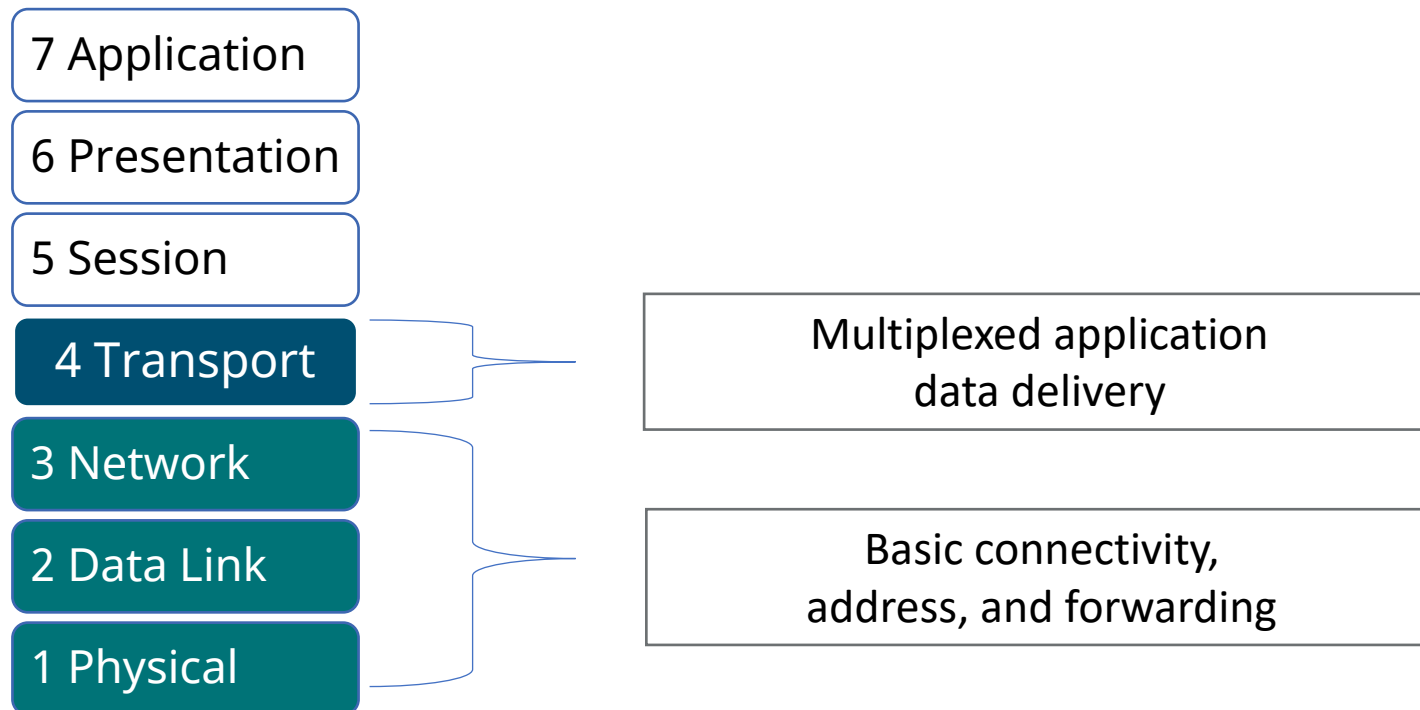
Learning Objectives

- Compare and contrast transport protocols.
- Use command line tools to scan network ports.
- Explain the use of network addressing services.
- Explain the use of name resolution services.
- Configure and troubleshoot DHCP and DNS services.

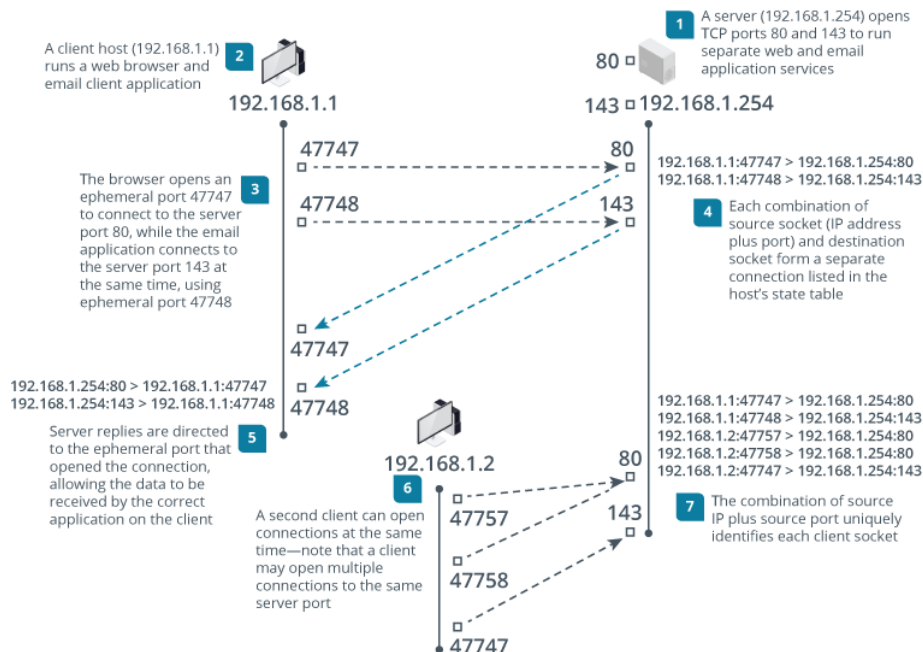
| Lesson [6.1]

Transport & Application Layer Protocols

Transport Layer



Transport Layer Ports and Connections



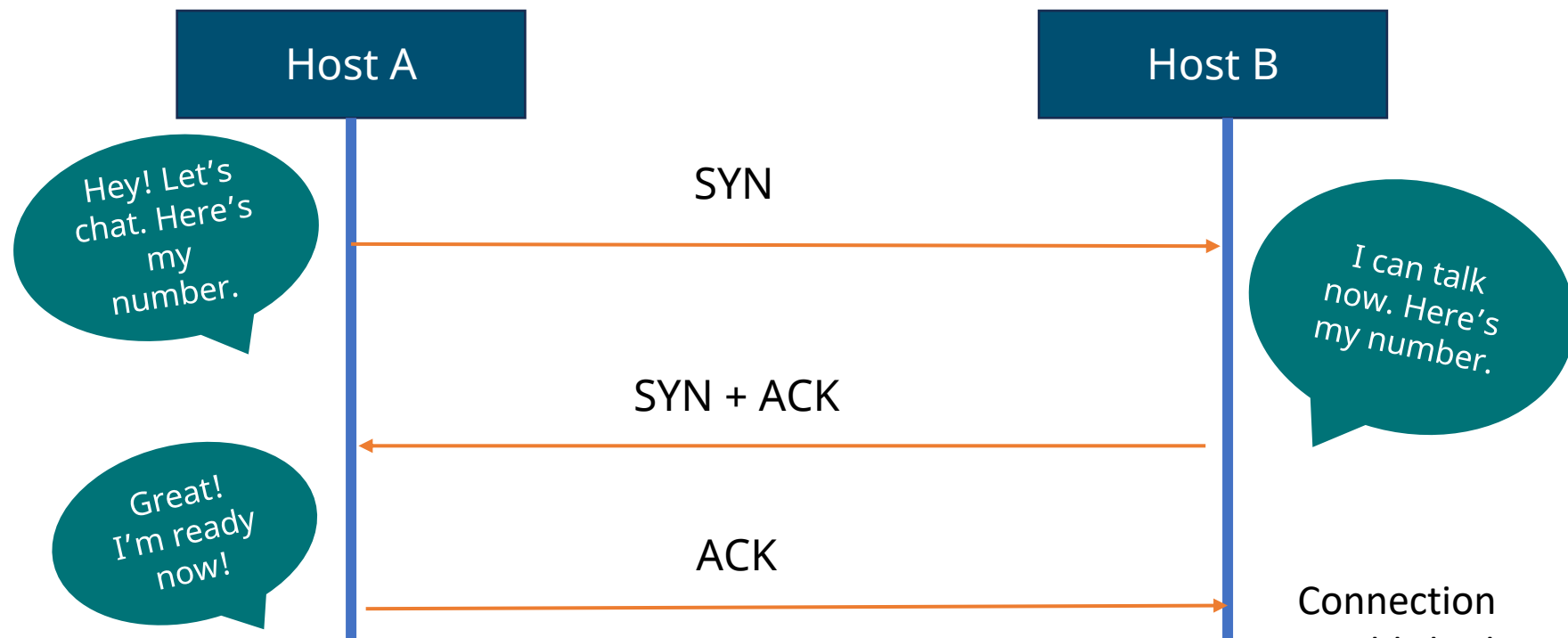
Transmission Control Protocol (TCP)

TCP helps computers send information to each other in a reliable way.

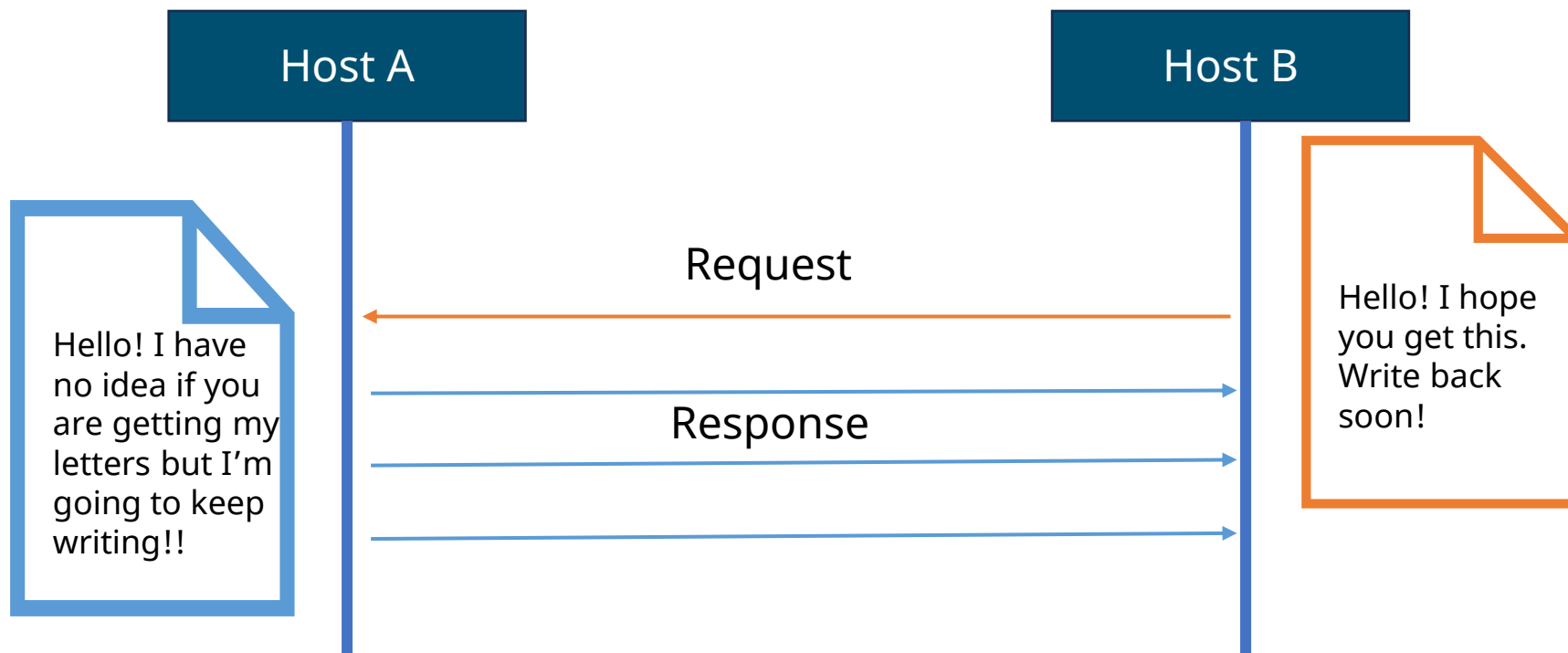
For reliability, TCP requires numerous header fields to maintain state information.

Source port
Destination port
Sequence number
Ack number
Data length
Flags
Window
Checksum
Urgent Pointer
Options

TCP Handshake



UDP Exchange



User Datagram Protocol (UDP)

UDP helps computers send information to each other in a non-guaranteed way.



Source port

Destination port

Message length

Checksum

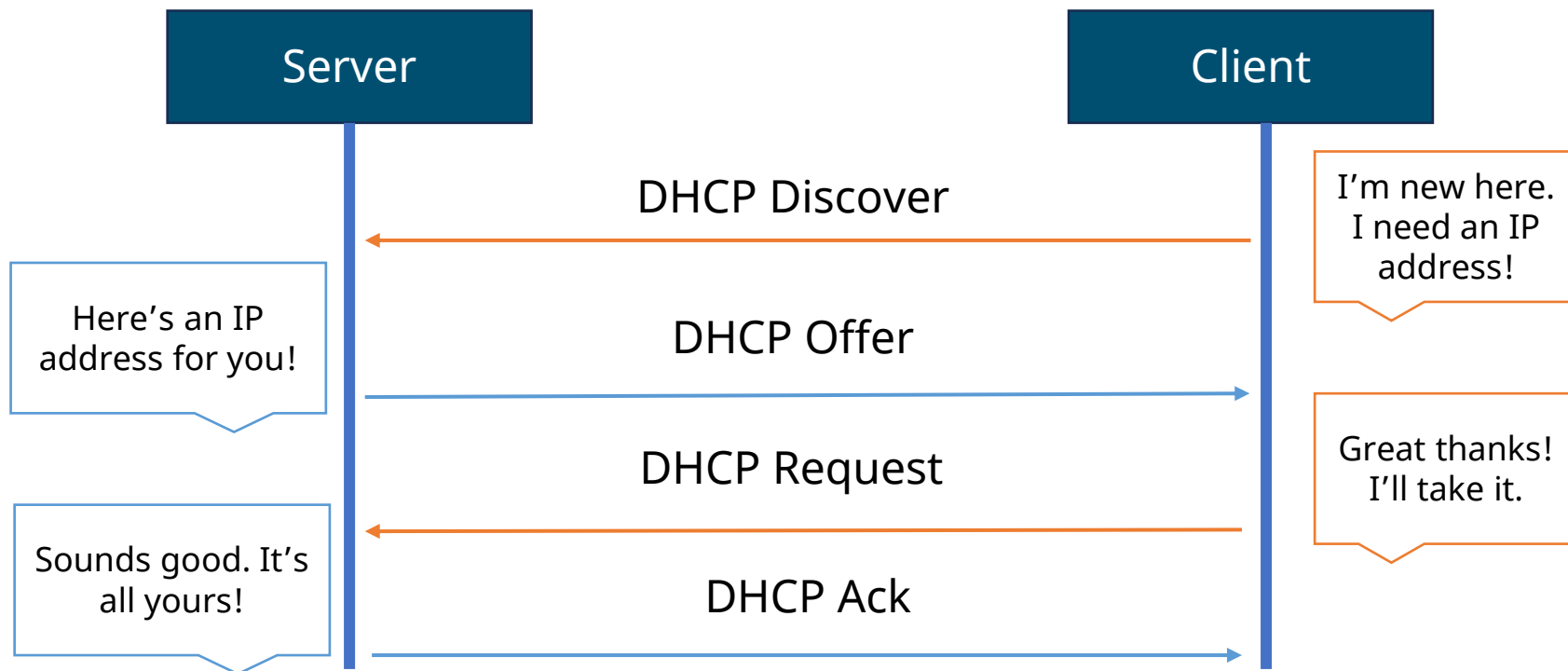
Activity: Run netstat

1. Open the command prompt or Terminal.
2. Run **netstat** command to view all active connections and their respective ports.
3. Use additional options like **-a** to show all connections or **-b** to display the application making the connection.

Lesson 6.2

DHCP

DHCP Process



DHCP Server Configuration

A DHCP server must be

- Assigned a static IP address
- Configured with a range of IP addresses/subnet masks (scope)



The screenshot shows a 'DHCP Settings' window with a green question mark icon in the top right corner. The settings are as follows:

Setting	Value	Notes
DHCP Server:	☐ Auto ☒ On ☐ Off	
IP Address Pool:	192.168.1.100 - 192.168.1.199	
Address Lease Time:	720	(1-2880) minutes
Default Gateway:	192.168.1.254	(Optional)
Primary DNS:	192.168.1.254	(Optional)
Secondary DNS:	0.0.0.0	(Optional)

A green 'Save' button is located at the bottom right of the window.

DHCP Reservations and Exclusions

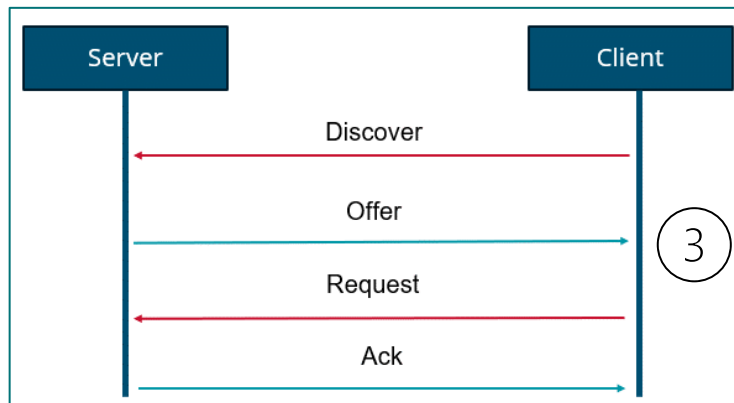
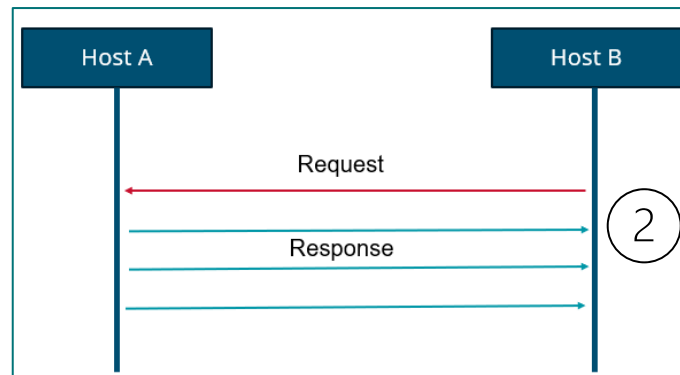
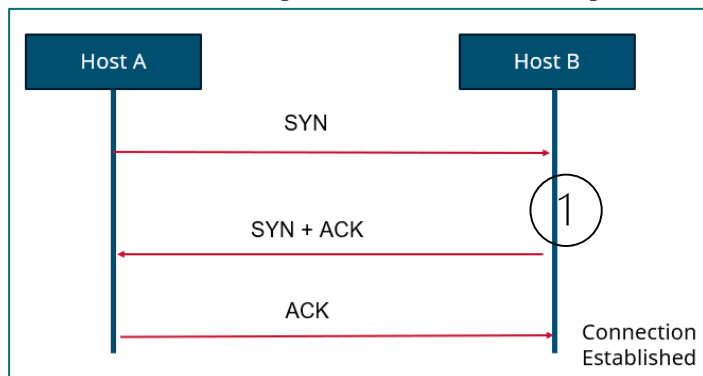
DHCP Reservations

- Ensures a device always receives the same IP address
- Tied to a device's MAC address
- Ideal for devices that need a consistent IP address

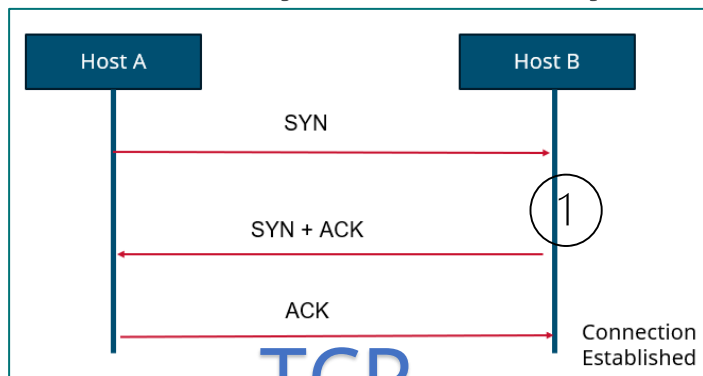
DHCP Exclusions

- Prevents a range of IP addresses from being assigned
- A range of IP addresses is set aside and not distributed
- Ideal for addresses statically assigned to devices

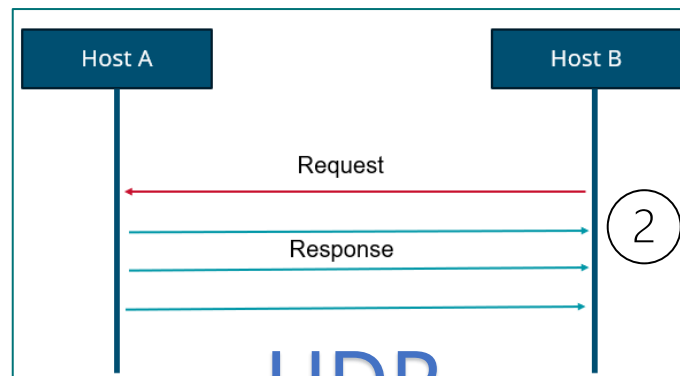
Activity: Identify the Protocol



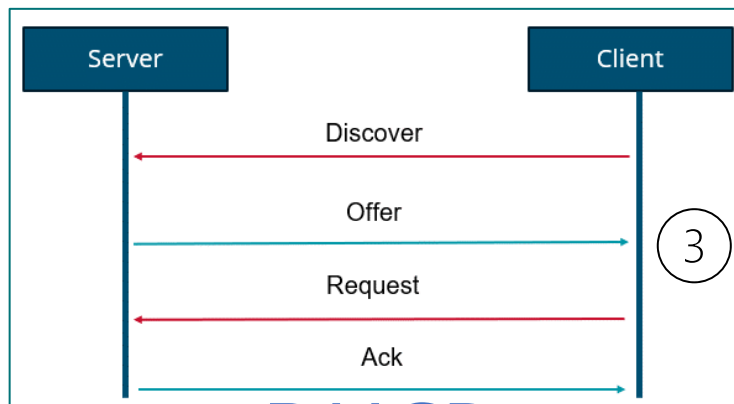
Activity: Identify the Protocol



TCP



UDP



DHCP

Lesson 6.3

APIPA and SLAAC

APIPA

Automatic Private IP Addressing

- Devices can self-assign an IP address when DHCP servers unavailable
- Only for local network use

How APIPA Works on a Device

- Assigns itself an address between 169.254.0.1 and 169.254.254.255
- Uses ARP to ensure address is unique

IPv6 Autoconfiguration

IPv6 Autoconfiguration Description

- Allows devices on an IPv6 network to configure their own IP addresses without DHCP

Types

- Stateless Address Autoconfiguration (SLAAC)
- Stateful Address Autoconfiguration (DHCPv6)

DHCPv6

Function

DHCPv6 request and response

Address assignment and management

Lease time and renewal

Advantages

Centralized control

Extensive network configuration options

Limitations

Requires dedicated server setup

More complex than SLAAC

Lesson 6.4

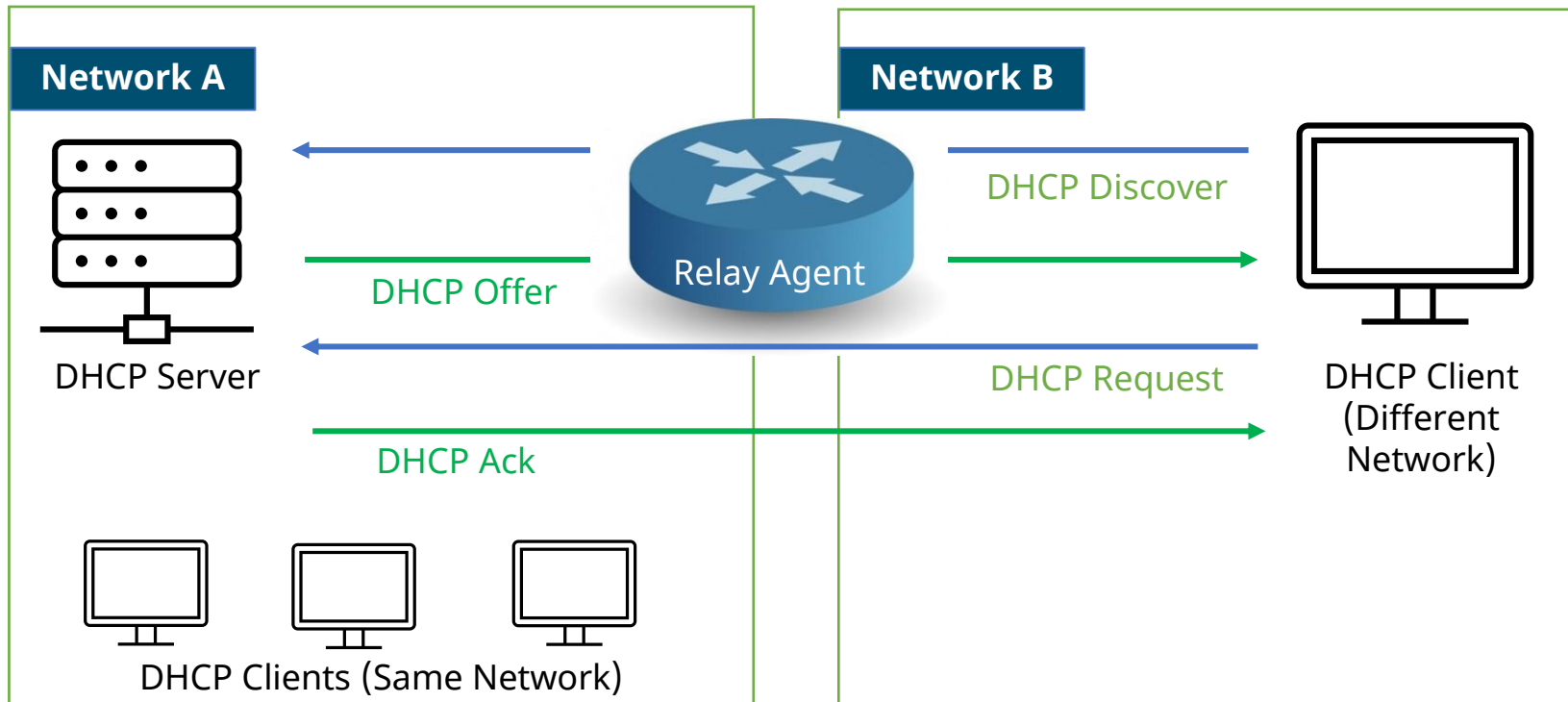
DHCP Relay and Troubleshooting

DHCP Relay

DHCP Relay Agent

- Acts as a network bridge
- Receives DHCP broadcast messages from clients and forwards them to DHCP server
- Enables clients to obtain IP address from remote DHCP server

DHCP Relay



IP Helper

IP Helper

- Cisco network device feature
- Forwards DHCP and other broadcast traffic across different subnets

Functionality

- Translates broadcasts to unicast to cross subnet boundaries
- Sends DHCP requests from clients to a server in a different subnet

Quiz: What is the possible reason that Client is unable to reach external networks

- a) Incorrect subnet mask
- b) Incorrect operating system
- c) Incorrect default gateway
- d) Incorrect user login

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- a) Use inclusion ranges
- b) Use exclusion ranges
- c) Use IP Helper
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Lesson 6.5

Domain Naming System

Think About It: DNS

What is DNS
(Domain Name System)?



Domain Name Service (DNS)

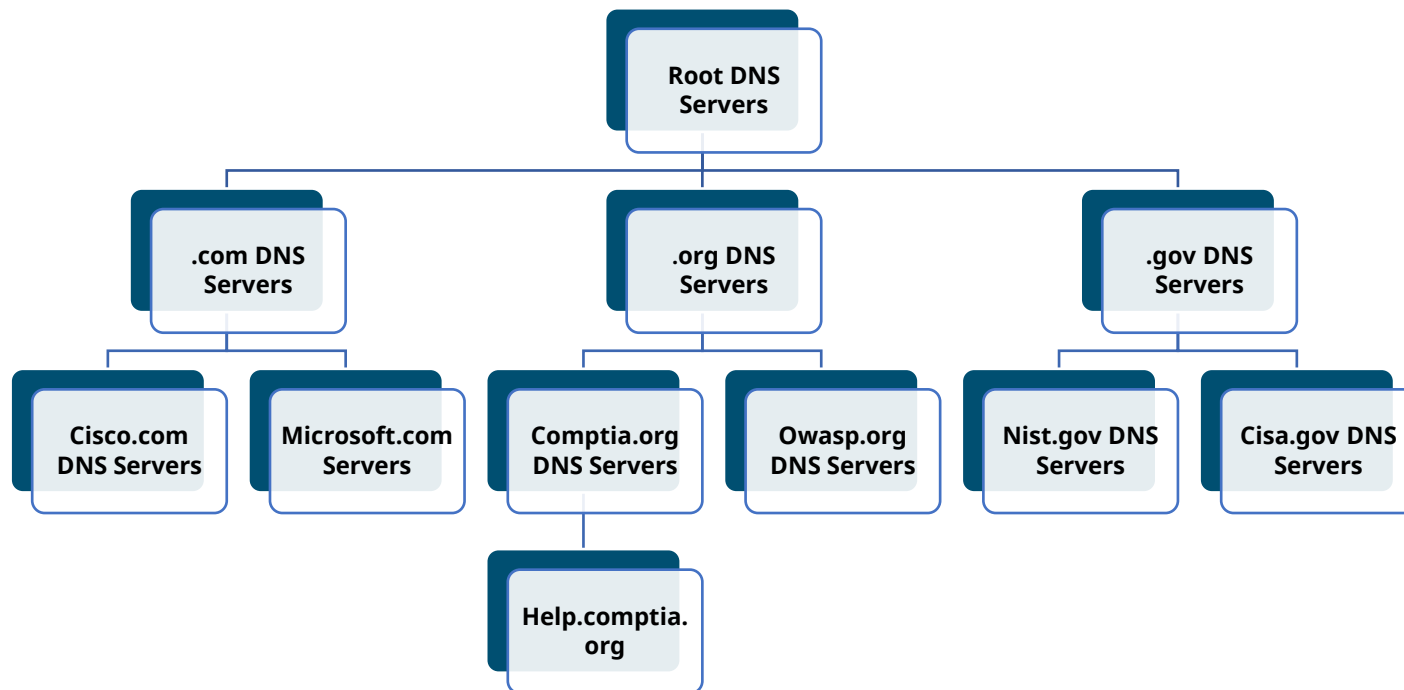
Host Names

- Local identifiers
- Assigned by administrator
- Human-friendly names
- Unique on the local network

Domain Names

- Global identifiers
- Hierarchical structure
- Avoids duplications on the internet
- Registered with a registrar

DNS Hierarchy



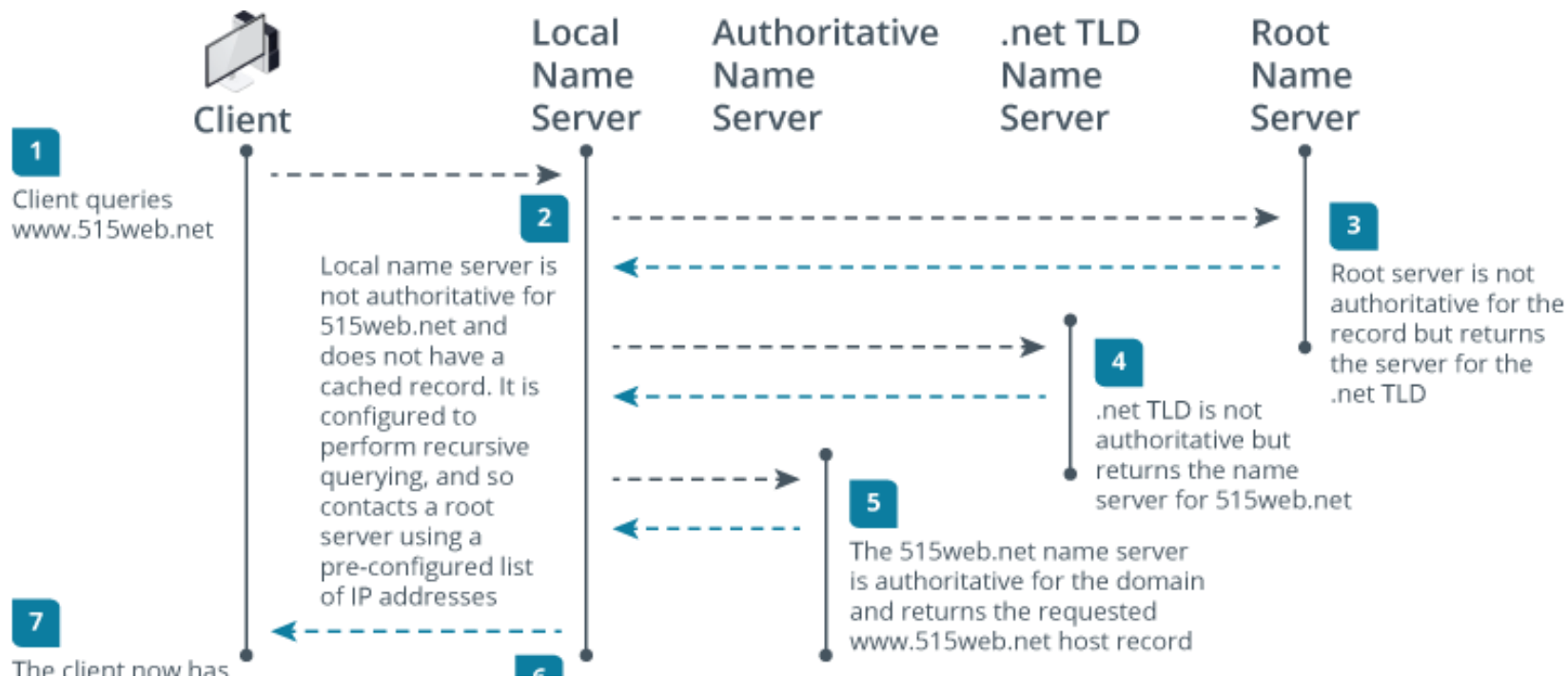
Root Domain

Top Level Domain

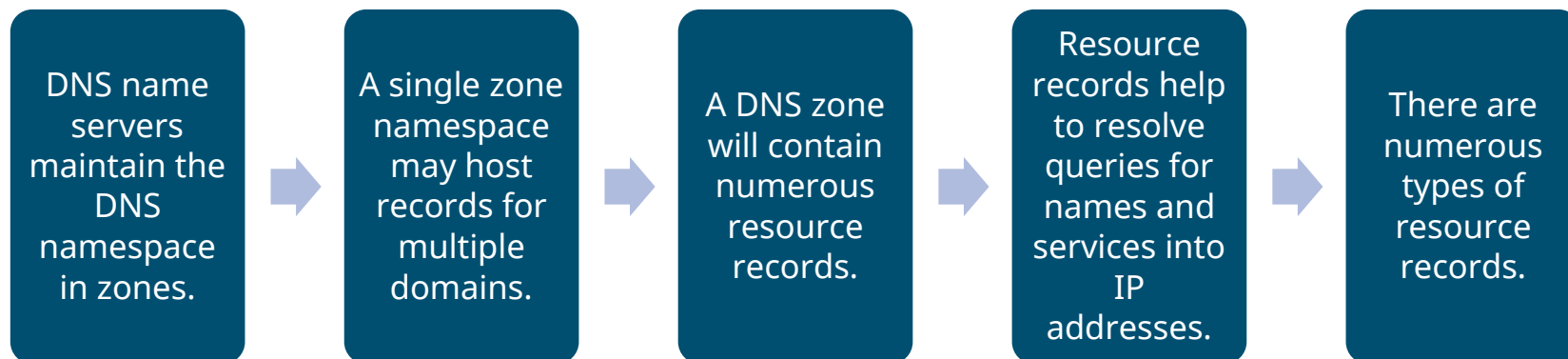
Domain

Sub-Domain

Name Resolution using DNS



Resource Records



Host Record Types

SOA Records

Indicate authority for a DNS zone

Contain admin contact, refresh rates, and versioning information

Host Address Records

Map domain names to IPv4 addresses

Needed for DNS to translate domain names into IP addresses

Mail Exchange Records

Specify mail servers responsible for a domain

Prioritize mail delivery

Service Records

Identify service locations within a domain

Include port numbers and priority for services

Text Records

Store text information for external sources

Used for SPF and DKIM records to reduce spam

Pointer Records

Used for reverse DNS lookup

Map IP addresses back to domain names

DNS Server Configuration



Primary DNS Servers

- Translate domain names into IP addresses
- Act as the first point of contact for DNS queries



Secondary DNS Servers

- Provide redundancy and load balancing for DNS queries
- Serve as backup servers if the primary server fails



Cache-only DNS servers

- Cache DNS resolutions to speed up response times
- Handle repetitive queries to reduce load on primary/secondary servers

Internal vs. External DNS

External DNS Servers:

- Public access: Resolves internet domain names for users
- External management: Managed by providers, not local network
- Public focus: Primarily handles public domain to IP resolution

Internal DNS Servers:

- Private network: Resolves internal domain names
- Internal control: Managed by IT for local network
- Security focus: Enhances network security by isolating traffic

DNS Security



Threat	Solution
DNS spoofing and poisoning	DNSSEC (DNS Security Extensions)
Eavesdropping and interception attacks	DNS over TLS (DoT)
Interception attacks and tampering	DNS over HTTPS (DoH)
DNS tunneling and advanced attacks	Advanced DNS Firewall

Activity: Two Truths and a Lie



Host names help to
avoid duplications on
the Internet

Resource records help
resolve queries for
names and services
into IP addresses

Pointer records map
IP addresses back to
domain names

Activity: Two Truths and a Lie

LIE

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Lesson 6.6

DNS Troubleshooting

Client DNS Issues

Symptoms of DNS Issues

- Timeouts and errors
- Partial DNS failures
- Page loading issues
- Delays in accessing websites



Name Resolution Issues

Name Resolution Methods

- Check local name caches
- Check HOSTS
- Verify DNS records using **nslookup** or **dig**



Activity: Run nslookup and dig

Windows Machines

- Open command prompt or Terminal
- Type **nslookup comptia.org**

Linux Machines

- Open Terminal
- Type **dig comptia.org**

Summary

- **TCP (Transport):** Reliable, connection-oriented, and error-checked data delivery (acknowledgments)
- **DHCP (Automatic IP):** Assigns IP addresses, subnet masks, and optional parameters (gateway, DNS).
- **Relay Agents (DHCP):** Forward DHCP traffic between subnets
- **DNS (Global Hierarchy):** Translates domain names to IP addresses.